COBRA Machine Learning

User Manual

Comprehensive Distance Based Ranking Integrated with Machine Learning, Gaussian Balancing and PSI

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Table of Contents

1. Introduction

COBRA Machine Learning (COBRA-ML) is a decision support software that combines the COBRA (Comprehensive Distance Based Ranking) multi-criteria method with Machine Learning techniques, Gaussian weight balancing, and the PSI (Preference Selection Index) method.

The tool is designed for researchers, analysts, and decision-makers who need to rank a set of alternatives based on multiple criteria, especially in contexts where:

- Criteria weights are not pre-defined or the user wants objective weight calculation
- Optimal parameter tuning is desired through automated learning
- The problem involves trade-offs between benefit criteria (maximize) and cost criteria (minimize)

COBRA-ML is a Streamlit-based web application, bilingual (Portuguese/English), running locally or on any server that supports Python.

1.1 Key Features

- Multi-criteria ranking using the COBRA method
- Objective weight calculation via PSI (Preference Selection Index)
- Gaussian balancing of weights with parameters lambda, mu, and sigma
- Automatic parameter prediction using a Random Forest Regressor
- Three built-in sample problems: Renewable Energy, Fighter Aircraft, and Suppliers
- Support for custom problems and file upload (Excel/CSV)
- Interactive visualizations (heatmaps, radar charts, Gaussian function plots)
- Export to CSV, Excel, and JSON
- Optional AI-powered analysis via OpenAI API
- Bilingual interface (Portuguese and English)

1.2 Application Areas

- Renewable energy source selection
- Military equipment acquisition and evaluation
- Supplier selection and procurement
- Technology assessment
- Any multi-criteria decision-making (MCDM) problem

2. Installation and Setup

2.1 System Requirements

Requirement	Version / Details
Python	3.8 or higher
Streamlit	1.x
scikit-learn	1.x
pandas	1.x or 2.x
numpy	1.x
plotly	5.x
openai (optional)	1.x (for AI features)

2.2 Installing Dependencies

Open a terminal and run:

```
pip install streamlit pandas numpy scikit-learn plotly openpyxl
xlsxwriter
```

To enable the optional AI analysis feature:

```
pip install openai python-dotenv
```

2.3 Running the Application

1. Save the file cobra_ml_app_integrated.py to your working directory.
2. Open a terminal in that directory.
3. Run the following command:
```
streamlit run cobra_ml_app_integrated.py
```
4. The application will automatically open in your default web browser at http://localhost:8501

Tip: If the browser does not open automatically, manually navigate to http://localhost:8501 in your browser.

3. Application Interface

3.1 Language Selection

At the top of the application, there is a language selector. You can switch between Portuguese and English at any time. The interface, labels, and output descriptions will update immediately.

3.2 Sidebar Settings

The left sidebar contains all configuration options:

Setting	Description
Data Source	Choose from built-in examples or provide your own data
Use Machine Learning	Enables ML-based prediction of optimal parameters (lambda, mu, sigma)
Train ML Model	Generates synthetic data and trains a new Random Forest model
Number of Training Samples	Controls how many synthetic problems are used for training (10-100)
OpenAI API Key	Optional — enables AI-powered result analysis

3.3 Data Sources

COBRA-ML offers five data input modes:

Mode	Description	When to Use
Renewable Energy	5 alternatives, 5 criteria (built-in)	Quick demo or energy sector studies
Fighter Aircraft	5 alternatives, 6 criteria (built-in)	Military or procurement case studies
Suppliers	4 alternatives, 5 criteria (built-in)	Supply chain and procurement analysis
Custom Problem	User-defined alternatives and criteria	Any new decision problem
Upload File	Excel (.xlsx) or CSV file	Pre-existing datasets

4. Step-by-Step Usage Guide

4.1 Using a Built-in Example

5. Select a Data Source in the sidebar (e.g., Renewable Energy).
6. The decision matrix will appear in the main panel — you can review and edit the values directly in the table.
7. Check the criteria types (Benefit = maximize, Cost = minimize) displayed below the table.
8. Optionally enable Machine Learning in the sidebar.
9. Click the Run COBRA-ML Analysis button.
10. View the results in the tabs: Final Ranking, Weights Analysis, Matrices, ML Parameters, and Visualizations.

4.2 Using a Custom Problem

11. Select Custom Problem in the Data Source dropdown.
12. Set the Number of Alternatives and Number of Criteria using the sliders.
13. Enter the names of your alternatives in the text fields.
14. Enter criterion names and select their type (Benefit or Cost) for each.
15. Fill in the decision matrix values in the editable table.
16. Click Run COBRA-ML Analysis.

Benefit criteria are those you want to maximize (e.g., quality, efficiency). Cost criteria are those you want to minimize (e.g., price, emissions).

4.3 Uploading a File

Your file must follow this format:

- First column: alternative names (labeled Alternatives or similar)
- Remaining columns: criteria values, one column per criterion
- First row: headers (column names)

Example Excel structure:

Alternatives	Cost ($)	Quality (1-10)	Delivery (days)
Option A	1000	8	15
Option B	1200	9	10
Option C	900	7	20

After uploading, configure the criteria types (Benefit/Cost) in the interface before running the analysis.

4.4 Configuring Machine Learning Options

Three modes are available for parameter configuration:

Mode	Description
ML with trained model	The app uses an existing trained model to predict lambda, mu, and sigma based on your problem features.
ML + Train new model	Generates synthetic training data and retrains the Random Forest model before predicting parameters.
Manual parameters	The user defines lambda, mu, and sigma manually via sliders in the Manual Parameters section.

5. Methodology

5.1 Overview of COBRA-ML

COBRA-ML combines four components into a single decision pipeline:

Component	Role
COBRA	Core ranking method using distances to Positive Ideal Solution (PIS) and Negative Ideal Solution (NIS)
PSI (Preference Selection Index)	Objective weight calculation based on variance of normalized criteria values
Gaussian Balancing	Adjusts weights using a Gaussian (bell-curve) function parametrized by lambda, mu, and sigma
Random Forest Regressor	Predicts optimal lambda, mu, and sigma from problem features automatically

5.2 Method Execution Flow

When you click Run COBRA-ML Analysis, the following steps are executed:

17. Decision Matrix Normalization: Each criterion column is normalized using vector normalization. Benefit criteria use the column vector; cost criteria use the inverse values.
18. PSI Weights Calculation: The Preference Selection Index method computes objective weights from the variance of normalized values. Criteria with higher variability across alternatives receive higher weight.
19. Parameter Prediction or Manual Input: If ML is enabled, the Random Forest model predicts the Gaussian parameters (lambda, mu, sigma) from features extracted from the decision matrix. Otherwise, default or user-defined values are used.
20. Gaussian Weight Balancing: The final weights are computed by combining PSI weights with a Gaussian distribution function, controlled by lambda (blending), mu (center), and sigma (spread).
21. COBRA Score Calculation: The weighted matrix is computed, and for each alternative, distances to the Positive Ideal Solution (PIS) and Negative Ideal Solution (NIS) are calculated. The final COBRA score is D_minus / (D_plus + D_minus).
22. Ranking: Alternatives are ranked in descending order of their COBRA-ML scores. The alternative with the highest score is the best choice.

5.3 Gaussian Parameters Explained

Parameter	Symbol	Range	Effect
Lambda	lambda	0.0 to 1.0	Blending coefficient: 0 = pure PSI weights, 1 = pure Gaussian-PSI blend

Mu	mu	0 to n-1	Center of the Gaussian bell curve (index of most emphasized criterion)
Sigma	sigma	0.5 to n	Spread of the bell curve (how many criteria are emphasized)

6. Reading the Results

6.1 Final Ranking Tab

Displays a ranked table of all alternatives with their COBRA-ML scores. The alternative in Position 1 is the best according to the model and the given criteria.

> *Higher COBRA-ML scores indicate better alternatives. Scores range from 0 to 1.*

6.2 Weights Analysis Tab

Shows two sets of weights for each criterion:

- PSI Weights (Objective): derived purely from data variability, without subjective input.
- Final Weights (Balanced): the result after Gaussian balancing — these are the actual weights used in the scoring.

Bar charts visualize the weight distribution across criteria.

6.3 Matrices Tab

Displays the internal matrices computed during the analysis:

- Normalized Matrix (R): the decision matrix after normalization.
- Weighted Matrix (V): the normalized matrix multiplied by the final weights.

6.4 ML Parameters Tab

Shows the three key parameters used in the analysis:

Parameter	Description
lambda	Weight blending factor between PSI and Gaussian weighting
mu	Center of the Gaussian emphasis on criteria
sigma	Spread (bandwidth) of the Gaussian emphasis

If ML was used, a note confirms the parameters were predicted by the model. Otherwise, default or manually set values are shown.

6.5 Visualizations Tab

- Gaussian Balancing Function Plot: Shows the bell-curve distribution applied to criteria weights.

- Heatmap: Shows the normalized matrix as a color-coded grid, highlighting strong and weak performers per criterion.
- Radar Chart: Shows the normalized profile of a selected alternative across all criteria.

7. Exporting Results

After running an analysis, the Export Results section becomes available. Three formats are supported:

Format	Content	Use Case
CSV	Ranking, scores, PSI weights, final weights	Quick tabular analysis in Excel or Python
Excel (.xlsx)	Ranking sheet + Normalized Matrix + Weighted Matrix	Full structured report
JSON	Ranking, scores, model parameters, method label	Integration with other systems or further processing

Click the corresponding Download button to save the file locally.

8. AI-Powered Analysis (Optional)

COBRA-ML includes an optional integration with the OpenAI API (GPT-4o-mini model) for automated interpretation of results.

8.1 Configuration

23. Obtain an OpenAI API key from https://platform.openai.com
24. Enter the key in the OpenAI API Key field in the sidebar.
25. Alternatively, create a file named code.env in the application directory with the content:
    ```
    OPENAI_API_KEY=your_key_here
    ```

8.2 Using AI Analysis

26. After running a COBRA-ML analysis, scroll to the AI Analysis section.
27. Click Generate AI Analysis.
28. The AI will provide: result interpretation, weights analysis, evaluation of the winning alternative, and recommendations.
29. Use the Interactive AI Chat to ask follow-up questions about the results.

> *AI analysis is optional and requires an active OpenAI API key. The AI receives only the ranking scores, weights, and parameters — no personal or sensitive data is transmitted.*

9. Input File Format Reference

9.1 CSV Format

The first row must contain column headers. The first column should be the alternative names:

```
Alternative,Criterion1,Criterion2,Criterion3
Alt_A,45,8,120
Alt_B,38,7,95
Alt_C,52,9,110
```

9.2 Excel Format

The structure follows the same rules as CSV. All data must be numeric (except the Alternatives column). Empty cells and non-numeric values may cause errors.

9.3 Common Errors and Solutions

Error	Cause	Solution
Non-numeric values in matrix	Text or empty cells in data columns	Ensure all criterion values are numbers
File not loading	Unsupported format	Use .csv or .xlsx only
Wrong criteria types	All criteria default to Benefit	Review and change criteria types after uploading
ML model not available	Model not trained yet	Enable Train ML Model in settings
OpenAI error	Invalid or missing API key	Check that the API key is correct and has available credits

10. How to Cite

If you use COBRA-ML in your research, please cite as follows:

APA Format

Software: Portella, A. G., Santos, M. dos, & Gomes, C. F. S. (2025). COBRA Machine Learning Tool [Computer software].

ABNT Format

Software: SANTOS, Marcos dos; GOMES, Carlos Francisco Simoes. COBRA Machine Learning Tool. [S.I.]: Anderson Goncalves Portella, 2025. Programa de Computador.

11. Developers and Contact

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COBRA-ML was developed in the context of research in Production Engineering and Operations Research at Universidade Federal Fluminense (UFF) and Escola Naval, Brazil.

Appendix: Built-in Sample Problems

A. Renewable Energy Selection

Alternative	LCOE ($/MWh)	Capacity Factor (%)	CO2 Emissions (g/kWh)	Area (km2/MW)	Tech Maturity (1-10)
Solar PV	45	25	50	4.0	9
Onshore Wind	38	35	12	0.5	9
Offshore Wind	65	45	10	0.3	7
Hydroelectric	55	50	24	15.0	10
Biomass	60	70	230	2.0	8

Criteria types: LCOE = Cost, Capacity Factor = Benefit, CO2 = Cost, Area = Cost, Tech Maturity = Benefit

B. Fighter Aircraft Selection

Alternative	Cost (M$)	Speed (Mach)	Range (km)	Payload (kg)	Stealth	Maint.
F-35A	80	1.6	2200	8000	10	6
F-15EX	90	2.5	3900	13000	3	8
Rafale	85	1.8	3700	9500	5	7
Eurofighter	95	2.0	2900	7500	4	6
Su-57	100	2.0	3500	10000	8	4

Criteria types: Unit Cost = Cost; all others = Benefit

C. Supplier Selection

Alternative	Price ($)	Quality (1-10)	Delivery (days)	Sustainability (1-10)	Reliability (1-10)
Supplier A	100	8	15	7	9
Supplier B	120	9	10	9	8
Supplier C	90	7	20	6	7
Supplier D	110	8.5	12	8	8.5

Criteria types: Price = Cost, Delivery = Cost; Quality, Sustainability, Reliability = Benefit